Matter
Matter can be solid, liquid, or gas. Most of the matter around you, from planets to animals, is composed of mixtures of different substances. Only a few substances exist naturally in completely pure form.

A frog is made of compounds and mixtures.

An ice cream is an impure substance—a mixture of many different ingredients.

A sandwich is a mixture of several substances.

Impure substances
If a substance is impure, it means that something has been mixed into it. For example, pure water consists of only hydrogen and oxygen. But tap water contains minerals, too, which makes it an impure substance. All mixtures are impure substances.

Mixtures
There are many different kinds of mixtures, depending on what substances are in the mix and how evenly they mix. The substances in a mixture are not bound together chemically, and can be separated. Rocks are solid mixtures of different minerals that have been pressed or heated together.

A leaf is a very complex uneven mixture.

Muddy water is a suspension: it may look evenly mixed at first, but the larger mud particles soon separate out.

Heterogeneous mixtures
In a heterogeneous mixture, particles of different substances are mixed unevenly. Examples are concrete (a mixture of sand, cement, and stone) and sand on a beach, which consists of tiny odd-sized particles of eroded rock, sea shells, and glass fragments.

Suspensions
Suspensions are liquids that contain small particles that do not dissolve. If they are shaken, they can appear evenly mixed for a short time, but then the particles separate out and you can see them with your naked eye.

Colloids
A colloid looks like an even mixture, but no particles have been completely dissolved. Milk, for example, consists of water and fat. The fat does not dissolve in water, but floats around in minute blobs that you cannot see without a microscope. A cloud is a colloid of tiny water droplets mixed in air.

STATES OF MATTER

Most substances exist as solids, liquids, or gases—or as mixtures of these three states of matter. The particles of which they are made (the atoms, molecules, or ions) are in constant motion. The particles of a solid vibrate but are held in place—that's why a solid is rigid and keeps its shape. In a liquid, the particles are still attracted to each other, but can move over each other, making it fluid. In a gas, the particles have broken free from each other, and move around at high speed.

Changing states of matter

With changes in temperature, and sometimes in pressure, one state can change into another. If it is warm, a solid ice cube melts into liquid water. If you boil the water, it turns into gaseous steam. When steam cools down, it turns back into a liquid, such as the tiny droplets of mist forming on a bathroom window. Only some substances, including candle wax, exist in all three states.

Liquid
In the heat of the flame, the wax melts, and the molecules can move over each other and flow.

Gases
Near the wick, the temperature is high enough to vaporize the liquid wax, forming a gas of molecules that can react with the air. This keeps the flame burning.

Solid
Solid wax is made of molecules held together. Each wax molecule is made of carbon and hydrogen atoms.

Plasma, the fourth state of matter

When gas heats up to a very high temperature, electrons break free from their atoms. The gas is now a mixture of positively charged ions and negatively charged electrons: a plasma. A lightning bolt is a tube of plasma because of the extremely high temperature inside it. In space, most of the gas that makes up the sun, and other stars in our universe, is so hot it is plasma.